

Assignment 2: Hidden Reward Grid – Maximum Reward Path Using Q-Learning

Problem Statement Design and implement a Q-Learning agent that learns to travel from a start position to a goal position while maximizing the total reward collected. Reward values are hidden from the agent and revealed only when a cell is visited.

Description

The grid contains positive rewards, negative rewards, a Start position (S), and a Goal position (G). The agent must learn which regions are beneficial and which are harmful while reaching the goal efficiently

Example Hidden Grid (Known Only to Simulator)

```
S  9 -6 -2  1
-9  2 -5  1  1
 1 10 -5  2 -2
-3  5  2  5 -1
-5 -2  4 10  2
-2  1  2  3  G
```

User Inputs

- ✓ Number of rows (N)
- ✓ Number of columns (M)
- ✓ Start position
- ✓ Goal position
- ✓ Integer value input (range (1 to 10) and (-1 to -10))
- ✓ Learning rate (α)
- ✓ Discount factor (γ)
- ✓ Exploration rate (ϵ)
- ✓ Number of training episodes

Action Space Left, Right, Down, Up

Reward Function

- Visit cell : add value to reward
- Valid Move: -5
- Reach Goal: +100
- Invalid Move: -10

What You Have to Find

1. Maximum reward path from Start to Goal
2. Total reward collected
3. Number of steps required
4. Optimal policy learned by Q-Learning

What You Have to Show(Display)

- Final Q-table
- Learned path
- Total reward per episode graph
- Best path discovered after training

Episode Termination

The agent reaches the Goal (G).

The agent exceeds the maximum allowed number of steps.

(MaxSteps=2×Rows×Columns)

Constraints

- The agent starts at S and must reach G.
- The agent cannot move outside the grid boundaries.
- Each grid cell can be visited only once during an episode (no repetition).
- A visited cell becomes unavailable for future moves in the same episode.
- A step penalty is applied for every move.